ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000007 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000022 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

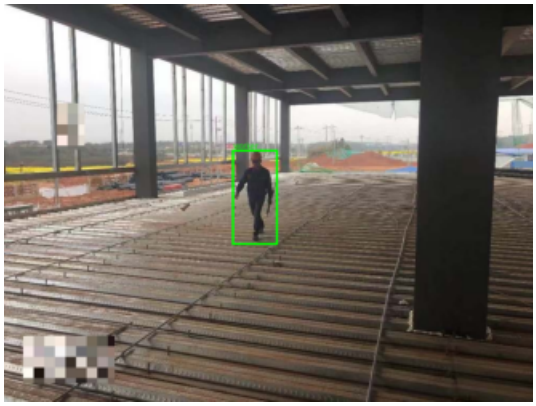
AFTER — region masked

img 0000023 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “hard hat”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined



AFTER — region masked

img 0000034 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000037 · rule_1 (hard hat) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000057 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000069 · rule_1 (hard hat) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000074 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000079 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000081 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000102 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

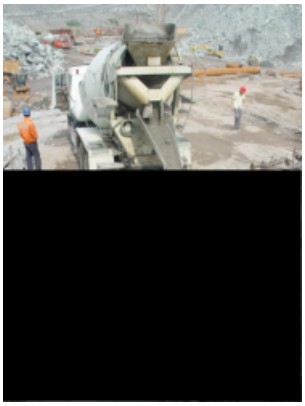
img 0000116 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked



img 0000117 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000120 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000124 · rule_2 (harness) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000129 · rule_1 (hard hat) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000149 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000183 · rule_1 (hard hat) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

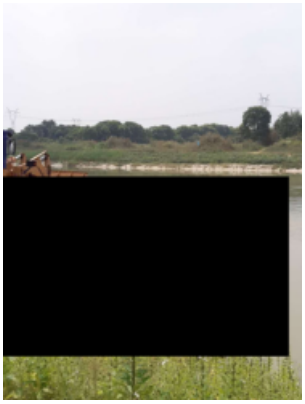
AFTER — region masked

img 0000203 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

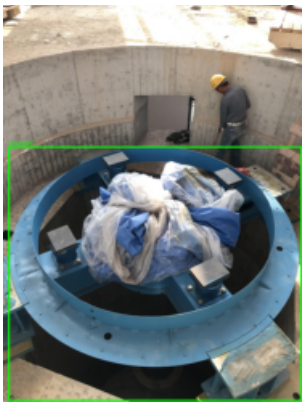
AFTER — region masked



img 0000204 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “hard hat”

BEFORE — region outlined



AFTER — region masked

img 0000205 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000229 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000233 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000253 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

NO BOX RETURNED → masked top-left cell of 4×4 grid fallback

commands (returned nothing): <OPEN_VOCABULARY_DETECTION>worker <OPEN_VOCABULARY_DETECTION>guardrail

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000254 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000280 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “harness”

BEFORE — region outlined

AFTER — region masked

img 0000288 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000325 · rule_4 (excavator) · GT: compliant · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000327 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000331 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000332 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000341 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000361 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

NO BOX RETURNED → masked top-left cell of 4×4 grid fallback

commands (returned nothing): <OPEN_VOCABULARY_DETECTION>worker <OPEN_VOCABULARY_DETECTION>guardrail

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000370 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000376 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000413 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “harness”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000421 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000433 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000437 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

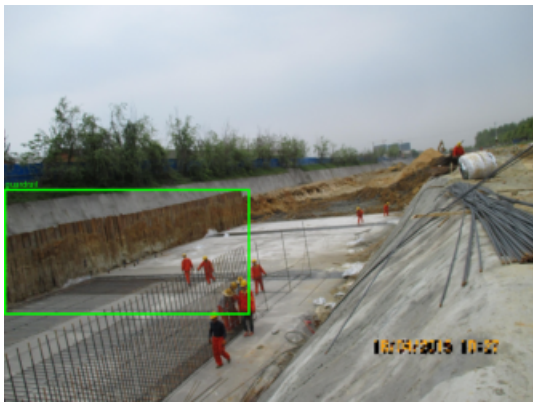
BEFORE — region outlined

AFTER — region masked

img 0000444 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “harness”

BEFORE — region outlined



AFTER — region masked

img 0000468 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

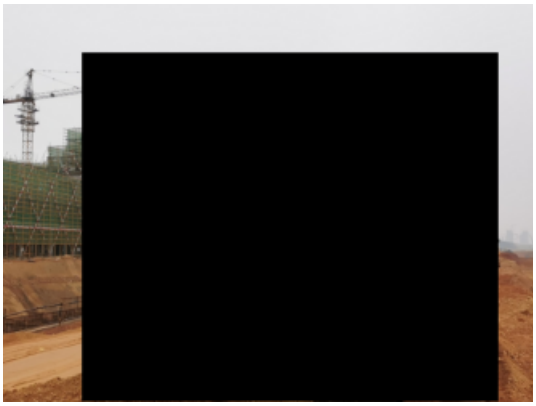
img 0000469 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked



img 0000471 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

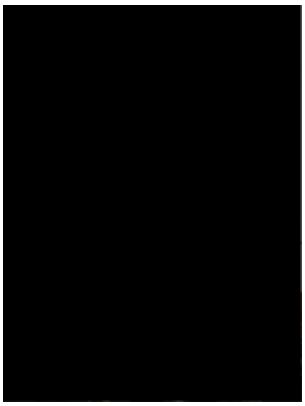
AFTER — region masked

img 0000478 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked



img 0000486 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000490 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000503 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000514 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000530 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000545 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

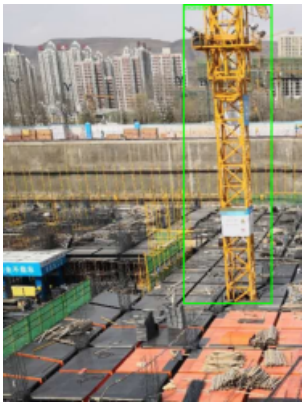
AFTER — region masked

img 0000550 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined



AFTER — region masked

img 0000551 · rule_4 (excavator) · GT: compliant · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000559 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000568 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000606 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

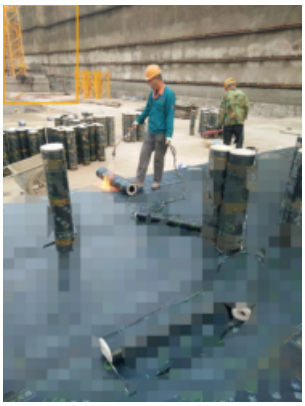
BEFORE — region outlined

AFTER — region masked

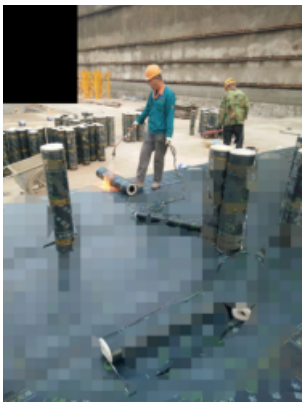
img 0000613 · rule_4 (excavator) · GT: compliant · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined



AFTER — region masked



img 0000620 · rule_2 (harness) · GT: compliant · pred: violation

NO BOX RETURNED → masked top-left cell of 4×4 grid fallback

commands (returned nothing): <OPEN_VOCABULARY_DETECTION>worker <OPEN_VOCABULARY_DETECTION>safety harness

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000623 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

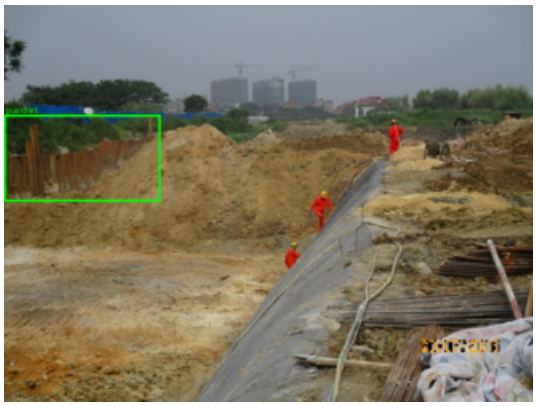
BEFORE — region outlined

AFTER — region masked

img 0000625 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined



AFTER — region masked

img 0000626 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000632 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000641 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000642 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000745 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000755 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000760 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000766 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000798 · rule_3 (guardrail) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000804 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “hard hat”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

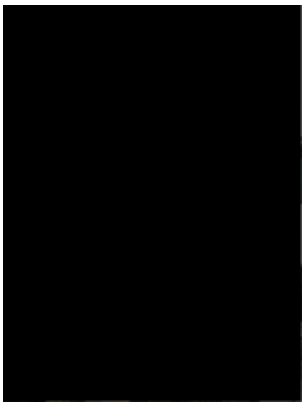
AFTER — region masked

img 0000874 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked



img 0000878 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0000879 · rule_1 (hard hat) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000886 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000888 · rule_4 (excavator) · GT: compliant · pred: safe

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0000889 · rule_4 (excavator) · GT: compliant · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000895 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000904 · rule_2 (harness) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “harness”

BEFORE — region outlined

AFTER — region masked

img 0000905 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000923 · rule_2 (harness) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “harness”

BEFORE — region outlined

AFTER — region masked

img 0000925 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

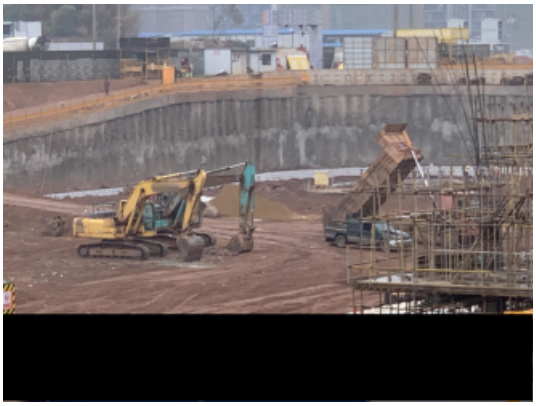
img 0000936 · rule_4 (excavator) · GT: compliant · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked



img 0000940 · rule_3 (guardrail) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0000959 · rule_2 (harness) · GT: compliant · pred: violation

top region: model box, ranked #1 by area · label = “harness”

BEFORE — region outlined

AFTER — region masked

img 0000991 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0000993 · rule_2 (harness) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001014 · rule_3 (guardrail) · GT: compliant · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001033 · rule_4 (excavator) · GT: struck_by_risk · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001071 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001077 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001081 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001117 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001151 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

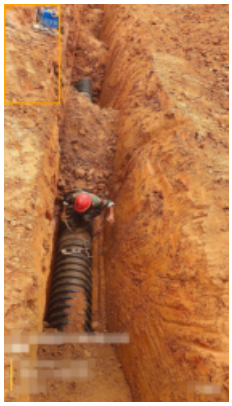
AFTER — region masked

img 0001194 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

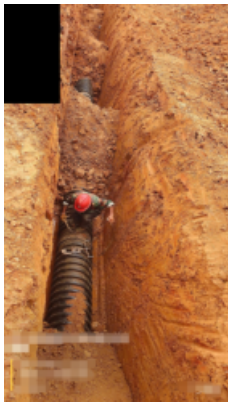
top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined



AFTER — region masked



img 0001229 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

NO BOX RETURNED → masked top-left cell of 4×4 grid fallback
commands (returned nothing): <OPEN_VOCABULARY_DETECTION>worker <OPEN_VOCABULARY_DETECTION>guardrail

BEFORE — region outlined

AFTER — region masked

img 0001274 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001325 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001431 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001436 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001460 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001470 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001484 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

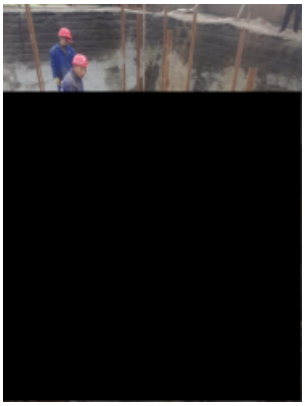
img 0001512 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked



img 0001520 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001530 · rule_2 (harness) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001597 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001623 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001667 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001699 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001798 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001804 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001842 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001981 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0001982 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0001986 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0001989 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0002042 · rule_4 (excavator) · GT: struck_by_risk · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0002092 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “harness”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0002150 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0002153 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0002158 · rule_2 (harness) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0002178 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0002211 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

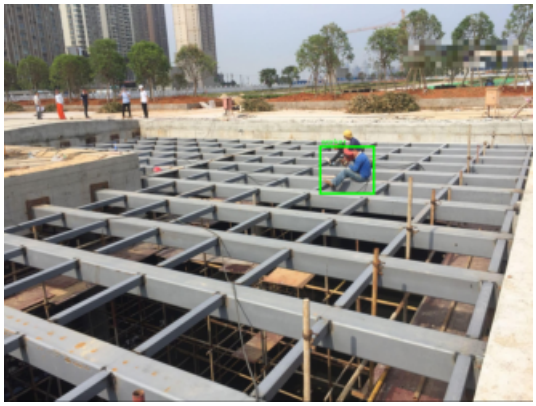
AFTER — region masked

img 0002260 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

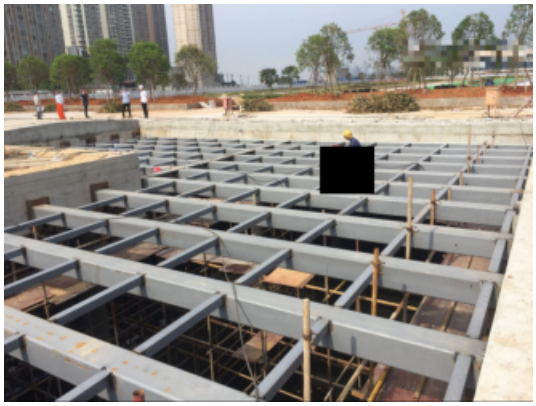
top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined



AFTER — region masked



img 0002262 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0002291 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0002348 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0002446 · rule_4 (excavator) · GT: struck_by_risk · pred: hazard

top region: model box, ranked #1 by area · label = “excavator”

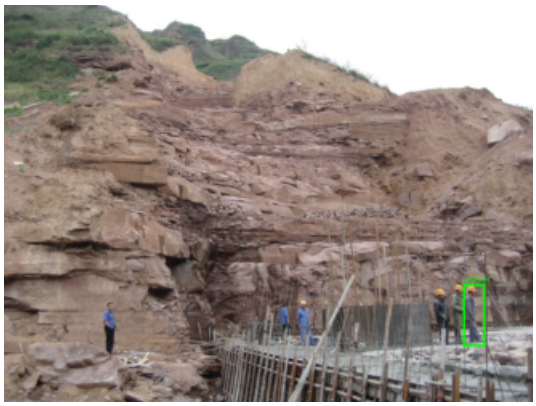
BEFORE — region outlined

AFTER — region masked

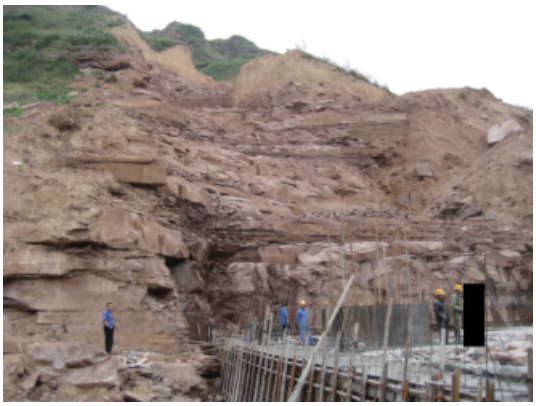
img 0002480 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined



AFTER — region masked



img 0002525 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0002534 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0002595 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0002599 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0002668 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0002815 · rule_2 (harness) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0002925 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0002957 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0003067 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0003071 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0003088 · rule_4 (excavator) · GT: struck_by_risk · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0003164 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0003206 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0003302 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0003339 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

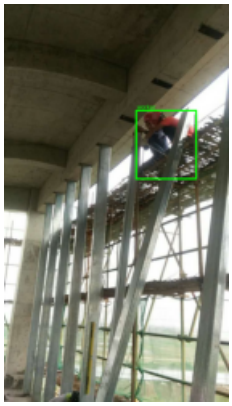
AFTER — region masked

img 0003429 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined



AFTER — region masked

img 0003479 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0003516 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0003613 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0003632 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “harness”

BEFORE — region outlined

AFTER — region masked

img 0003813 · rule_4 (excavator) · GT: struck_by_risk · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0003922 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “hard hat”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0003928 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

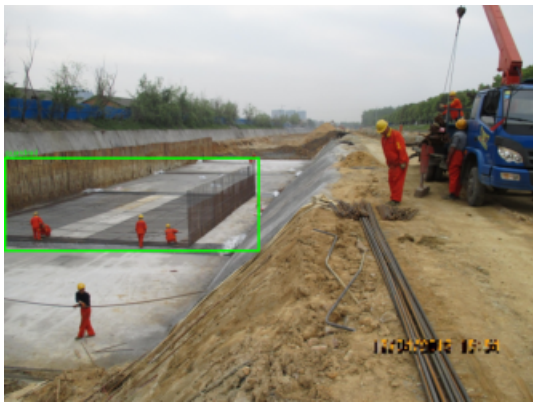
BEFORE — region outlined

AFTER — region masked

img 0003943 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined



AFTER — region masked

img 0004032 · rule_3 (guardrail) · GT: fall_hazard · pred: violation

top region: model box, ranked #1 by area · label = “guardrail”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0004171 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0004196 · rule_4 (excavator) · GT: struck_by_risk · pred: safe

top region: model box, ranked #1 by area · label = “excavator”

BEFORE — region outlined

AFTER — region masked

img 0004273 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0004332 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0004380 · rule_1 (hard hat) · GT: ppe_violation · pred: violation

top region: model box, ranked #1 by area · label = “worker”

BEFORE — region outlined

AFTER — region masked

img 0004497 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0004552 · rule_3 (guardrail) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “guardrail”

BEFORE — region outlined

AFTER — region masked

img 0004680 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “hard hat”

BEFORE — region outlined

AFTER — region masked

img 0004716 · rule_1 (hard hat) · GT: ppe_violation · pred: compliant

top region: model box, ranked #1 by area · label = “worker”

ConstructionSite XAI Pilot — Chapter 4: explanation-region masking (before → after)

BEFORE — region outlined

AFTER — region masked

img 0004831 · rule_2 (harness) · GT: fall_hazard · pred: compliant

top region: model box, ranked #1 by area · label = “worker”